



MIKROSKIL IDEAFUSE COMPETITION 2025

Fuel Frenzy

The **Kingdom of Mikroskil** faces an urgent crisis. Its grand energy reactor, which powers the capital, is on verge of collapse. Only a rare **Mikro Crystal** -- found deep within the mines of city -- can stabilize it.

The King has entrusted Oski, his most capable adventurer, with this vital mission: retrieve the crystal from City S and return to the Capital D .

Oski has successfully obtained the crystal, but time is running out. He needs to go back to the capital D from City S as fast as possible. Initially, his vehicle has F units of fuel and a full capacity of C units.

The kingdom consists of N cities and M bidirectional roads. The i -th road connects city U_i and city V_i , and it consumes f_i units of fuel and takes w_i time to traverse.

Each city i provides a gas station, allowing Oski to refill up to R_i units of fuel. However, each refueling action takes exactly T_i time, regardless of how much fuel he chooses to add.

Oski can refuel multiple times at the same city if needed, but each refueling action will again cost T_i time.

Determine the minimum total time needed for Oski to reach the capital D . If it's impossible, output -1 .

Input Format

$N M F C S D$

$U_1 V_1 f_1 w_1$

$U_2 V_2 f_2 w_2$

...

$U_M V_M f_M w_M$

$R_1 T_1$

$R_2 T_2$

...

$R_N T_N$

Output Format

The minimum total time for Oski to reach Capital S from City D with an initial fuel of F units. If it's impossible, output -1 instead.

Constraints

- $2 \leq N \leq 1000$
- $1 \leq M \leq \min(\frac{N(N-1)}{2}, 10^4)$
- $1 \leq C \leq 1000$

- $0 \leq F \leq C$
- $1 \leq S, D \leq N$ and $S \neq D$
- $1 \leq U_i, V_i \leq N$ and $U_i \neq V_i$
- $(U_i, V_i) \neq (U_j, V_j)$ if $i \neq j$
- $1 \leq f_i, R_i \leq C$
- $1 \leq w_i, T_i \leq 10^9$

Examples

Sample Input 1

```
4 4 3 5 1 4
1 2 3 5
2 3 2 5
3 4 5 8
4 1 5 30
2 5
1 2
5 2
4 2
```

Sample Output 1

```
24
```

Explanation

Oski starts at City 1 with 3 units of fuel and must reach City 4. Although he can travel directly from City 1 to City 4, that is not the best route to follow. Instead, he chooses the faster path that passes City 2 and City 3.

His journey goes as follows:

- Travel from City 1 to City 2 with 3 units of fuel. This takes 5 units of time.
- Refuel at City 2 to gain 1 unit of fuel. This takes 2 units of time.
- Refuel at City 2 again to gain another 1 unit of fuel. This takes another 2 units of time.
- Travel from City 2 to City 3 with 2 units of fuel. This takes 5 units of time.
- Refuel at City 3 to gain 5 units of fuel. This takes 2 units of time.
- Travel from City 3 to City 4 which costs 5 units of fuel. This takes 8 units of time.

No extra refueling is needed after that. Therefore, the minimum total time required is 24.

Sample Input 2

```
3 1 2 6 2 3
1 2 2 10
4 5
5 5
6 5
```

Sample Output 2

```
-1
```

Explanation

There is no path from City 2 to City 3. Thus, Oski can't reach his destination.